Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer pursuing an MSCS at USC with 2+ years shipping production applications and AI systems. Skilled in **TypeScript**, React, Node.js, Python, APIs, developer tooling, distributed workflows, and end-to-end ownership.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Software Engineering, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Web Technologies, Machine Learning

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant AI workflows with explicit job states, reusable service interfaces, and background workers, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **platform reliability** across 5K+ monthly tasks.
- Owned multi-tenant application improvements for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through **performance debugging**, indexed queries, code reviews, and independently shipped releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built React interfaces and Flask services for uploads, grading history, and feedback, supporting 150+ instructors and 10K+ daily requests through **full-stack ownership**, REST APIs, and production monitoring.
- Developed concurrent AI processing with provider abstractions, model switching, and failure isolation, handling 1K+ files per batch while **automated testing and CI/CD** reliably reduced feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed reusable integration modules with object-oriented patterns, improving API response times 30%, reducing technical debt 20%, and strengthening **developer adoption** across distributed Windows, Linux, and macOS environments.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **code review**, defect diagnosis, stakeholder feedback, and independent releases across three operating systems.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight TypeScript and Node.js services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ distributed records through **shared interfaces** while reducing staged production conflicts 33% during releases.
- Developed React and TypeScript workflows for real-time records using WebSockets and REST integrations, while Jenkins CI/CD and Docker staging reduced **deployment time** 50% across controlled production releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built React and TypeScript booking flows for scheduling, reservations, pricing, and payment status across 50K+ records, delivering sub-200-millisecond **full-stack features** for successful client demonstrations and stakeholder reviews.
- Designed modular interfaces for 12+ integrations, aligning REST APIs, Firebase synchronization, webhooks, and cross-platform delivery while reducing **design errors** 25% through early technical feedback and collaborative planning.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, C++, HTML5, CSS3, Bash
Frontend & Backend: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Shared Packages, SDK Design, CLI Tools, OpenAPI, Microservices, Distributed Systems, Asynchronous Processing
Data & AI: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, SQLite, LLM APIs, AI Workflows, Prompt Engineering, Vector Search, Data Modeling
Testing & Delivery: Git, GitHub, Code Review, Vitest, Jest, PyTest, Unit Testing, Integration Testing, CI/CD, GitHub Actions, Jenkins, Docker, Kubernetes, AWS, Azure, Vercel, Agile/Scrum, Debugging, Observability

Projects

DevBridge SDK | *TypeScript, Node.js, npm, OpenAPI, Vitest, GitHub Actions*

- Created a **developer-facing TypeScript SDK** and CLI that generate typed API clients from OpenAPI specifications, standardize retries and errors, and publish versioned npm packages through automated validation workflows.

Chat2SQL | *Java, TypeScript, Node.js, Docker, PostgreSQL, SQL Server*

- Extended a cross-platform **developer workspace** supporting 10+ databases, combining TypeScript interfaces with schema exploration, query generation, Docker tooling, and reproducible local environments across Windows, macOS, and Linux.

FinNexus Live | *Next.js, TypeScript, Node.js, MongoDB, Plaid, WebSockets*

- Built an end-to-end **TypeScript platform** supporting 50K+ users and 1M+ transactions, integrating secure bank linking, real-time synchronization, advanced filtering, and containerized deployment across client and server components.